

1 Function Manual

Collaborator: Iverson Scarlett

Role: Backend Engineer | Azure AI & FastAPI Specialist

Project: *Give What You Ya No Uses* – AI and Azure-powered Circular Economy Platform

2 1. FastAPI + Azure AI Integration

Main Objective: Implement intelligent REST endpoints to automate key platform workflows using Azure AI.

Tasks:

- Create FastAPI endpoints that interact with:
 - **Azure OpenAI** for text generation (e.g., title and description suggestions).
 - **Azure Translator** for automatic translation of listings and chats.
 - **Azure Computer Vision** for image analysis (detecting item type and condition).
 - Package services using **Docker Compose**.
 - Deploy to **Azure Container Instances (ACI)** or **Azure Kubernetes Service (AKS)**, depending on availability.
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3 2. Smart Agent / Copilot Modules

Main Objective: Provide automated assistance to users during the item listing process.

Tasks:

- Design and train a Python agent to:
 - Automatically classify listings (category + condition).
 - Suggest shipping or donation locations (use dummy data for now).
 - Integrate a “listing assistant” using **GPT prompts (Azure OpenAI)** to:
 - Ask guided questions to users.
 - Generate optimized titles and descriptions from keywords.
 - Briefly document the conversational flow in **Markdown or YAML** (for future integration into LangChain or LangGraph).
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4 3. Content Moderation and User Roles

Main Objective: Protect the platform from prohibited or sensitive content and ensure system security via role-based authentication.

Tasks:

- Implement hybrid moderation filters (Computer Vision + NLP) to detect:
 - Inappropriate images.
 - Offensive language, sensitive content, or banned products in text.
 - Configure user roles in **FastAPI**:
 - Basic user
 - Moderator
 - Administrator
 - Implement **JWT with OAuth2** for authentication.
 - Log moderation activities (audit logs).
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5 4. Payment System Integration

Main Objective: Offer premium features through secure payments.

Tasks:

- Integrate **Stripe or PayPal SDKs** (either is acceptable) into the backend.
 - Can use the existing PHP backend or a separate Python microservice.
 - Handle **webhooks** to confirm payments and activate premium listings.
 - Use **Azure Key Vault** to securely store API tokens.
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6 5. Cloud DevOps

Main Objective: Establish a reliable and agile deployment and maintenance environment.

Tasks:

- Create **CI/CD pipelines using GitHub Actions** for automated deployments to Azure.
 - Set up basic monitoring:
 - **Azure Monitor** or **Prometheus + Grafana** (based on preference).
 - Alerts for high resource usage or critical errors.
 - Document the pipeline and deployment flow in the **project's README**.
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7 Email Verification for User Accounts

Main Objective: Ensure users register in a legitimate and secure way.

Tasks:

- Implement **email verification** (send confirmation link upon registration).

- Ensure **one account per email address**.
 - Store **verification status** in the user database.
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8 Availability & Coordination

Time Zone: Eastern Time (EST)

Available Hours:

- **Monday to Friday:** 6:00 p.m. – 10:00 p.m.
- **Weekends:** Full-time availability

Communication Channels:

- **Discord:** Team coordination and task reviews.
- **Email:** Technical follow-up and file sharing.
- **GitHub:** Issue tracking and pull requests.